

Li-Fi Based Automatic Traffic Signal Control for Emergency Vehicles

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Abstract—India is one of the fastest developing country in the world. One of the disadvantages of this unprecedented growth is the ineffective traffic management. Light Fidelity (Li-Fi) is a form of bi-directional Visible Light Communication (VLC) in which light is modulated at speeds greater than a human eye can follow. Li-Fi can be used to implement Vehicle to Vehicle (V2V) communication as it has many advantages over other communication protocols. One of the main advantages of Li-Fi is that it provides connectivity within a very large area with more security and higher data rates. Emergency vehicles such as Ambulances, Firefighting trucks, Police vehicles etc. can propagate faster through traffic-dense roads using Li-Fi based V2V communication system. In this paper, a V2V communication system is proposed in which the headlight and tail light of a vehicle are made as Li-Fi transmitter and Li-Fi receiver respectively. Using the proposed system, when a traffic signal receives an alert regarding an emergency vehicle, the traffic signal would immediately turn green. Thus, decreasing the waiting time of emergency vehicles in traffic dense lanes. The proposed system was designed, implemented and tested for its operation.

Keywords—Li-Fi, V2V, VLC, Traffic management

I. INTRODUCTION

Wireless communication has been solely based on radio waves. Due to technological developments, this spectrum has become congested. Similarly, the traffic density has also increased many folds. To have an efficient traffic management, we need reliable wireless V2V communication system. The choice of radio waves for this application involves a huge bandwidth to be allocated to implement V2V communication system, which is very difficult as this spectrum is already congested due to human cellular communication systems [1]. Also, if we consider the circuitry associated with this type of communication, it is expensive. Therefore, the need for an alternative spectrum is increasing day by day. Gamma rays, X-rays, Infra-red and Ultraviolet rays cannot be used to implement V2V communication system due to their harmful effects on human body.

VLC is an emerging communication platform which uses visible light in the frequency range of 400 to 800 THz.

Li-Fi is a VLC protocol, which is like the Wi-Fi protocol in every layer except the physical layer [2]. The concept of Li-Fi

was introduced by Dr. Harald Haas during his speech at TED Global 2011[3]. He considered LED bulbs as the principle source of light because older sources such as incandescent and fluorescent light bulbs have proved themselves to be inefficient in terms of luminosity and energy consumption when compared to LEDs. Their intensity can be modulated at very high speeds. This property is exploited in Li-Fi technology. Li-Fi also enables transmission of thousands of data streams in parallel at high speeds. A photodiode is used as the receiving element in this technology. Both these devices are cheap to acquire, which makes Li-Fi a economically viable solution to all the problems offered by radio wave communication.

In India, many lives have been lost as emergency vehicles like ambulance or firefighting vehicles were not able to reach their destination on time due to heavy traffic. An efficient and feasible traffic management system can be implemented using Li-Fi [9]. In this paper, a Li-Fi based emergency vehicle detection and traffic clearance system has been implemented.

The paper is organized as follows: Section II gives the literature survey performed, Section III explains the methodology, Section IV presents the implementation, Sections V and VI discuss the advantages and disadvantages of using Li-Fi protocol for this application. Section VII provides the implementation results and Section VIII concludes the paper.

II. LITERATURE SURVEY

Research on traffic management, V2V communication, Ambulance based traffic control have been carried out since 1990s. Various communication protocols like GSM, RFID, IR, ZigBee etc. have been considered to implement emergency vehicle based traffic control. However, the above methods cannot be used in today's world for various reasons.

GSM based communication systems has been extensively used for cellular communication. The infrastructure needed for GSM communication is high compared to other communication protocols. Even though basic infrastructure for GSM is available, the use of this technology for V2V communication requires high bandwidth. RFID based V2V

communication do not require line of sight [4]. RFID based traffic control can detect emergency vehicles if it is in the frequency range of the traffic control. The prime disadvantage of RFID based V2V communication is low security offered where RFID messages are prone to unauthorized access. The metallic chassis of the vehicle can also cause problems to the RFID transceiver. IR based emergency vehicle detection is unreliable as IR signals require line of sight and sunlight can cause interference with IR signals. ZigBee based traffic management is a good alternative when compared to other communication protocols mentioned above. It offers wide range of communication, does not require line of sight and is comparatively economical. However, to implement V2V communication or traffic management using ZigBee involves restrictions on the number of nodes per controller [8].

Various research papers have proposed the use of Li-Fi for emergency vehicle based traffic control systems [5,6,7]. It does not need any infrastructure like GSM. It offers a much secure communication compared to RFID. It is relatively economical compared to ZigBee. It does not use the congested 2.4 GHz frequency like ZigBee. In this paper, Li-Fi based V2V communication system is implemented for smart control of traffic signal.

III. METHODOLOGY

A. Working Principle of Li-Fi

The elementary way of representing data in Li-Fi protocol is by making the intensity of the LED high, for a binary 1 and making the intensity of the LED low, for a binary 0. The variation in intensity will not affect the primary functionality of the LED fixture.

Any commercially available LED fixture along with some signal processing circuit can act as Li-Fi transmitter. Similarly, a photodiode along with a demodulation circuit can act as a Li-Fi receiver.

B. Modulation techniques used in Li-Fi

Various modulation techniques can be applied to transmit data using Li-Fi protocol. The modulation techniques should follow intensity modulation and direct detection (IM/DD) type of modulation. Some of them being ON-OFF keying(OOK), Pulse position modulation(PPM), orthogonal frequency division multiplexing(OFDM), Spatial OFDM, superposition OFDM etc.

Modulation techniques like OOK, PPM etc. which can be categorized as single carrier modulation do not support parallel transmission of data streams. On the other hand, OFDM based multiple carrier modulation techniques can be used for parallel transmission of data streams.

Transmission of data streams in parallel is mainly used for transferring images, videos or any other large data files. For

V2V communication where the data size is very small, simple modulation techniques are sufficient.

In this paper, ON-OFF keying modulation technique is used to implement V2V communication using Li-Fi.

C. Block Diagram

Fig. 1, shows the block diagram of the scenario considered to implement a traffic management system using Li-Fi. The diagram shows three main nodes namely ambulance (as an emergency vehicle), a non-emergency vehicle and a traffic signal control. The ambulance consists of a PIC microcontroller and a Li-Fi transmitter. The non-emergency vehicle consists of a PIC microcontroller along with a Li-Fi transceiver and the traffic light control consists of a PIC microcontroller along with a Li-Fi receiver. The Li-Fi transmitter for the non-emergency vehicle and ambulance can be implemented using the vehicle's headlights and the Li-Fi receiver can be fixed somewhere near the vehicle tail lights.

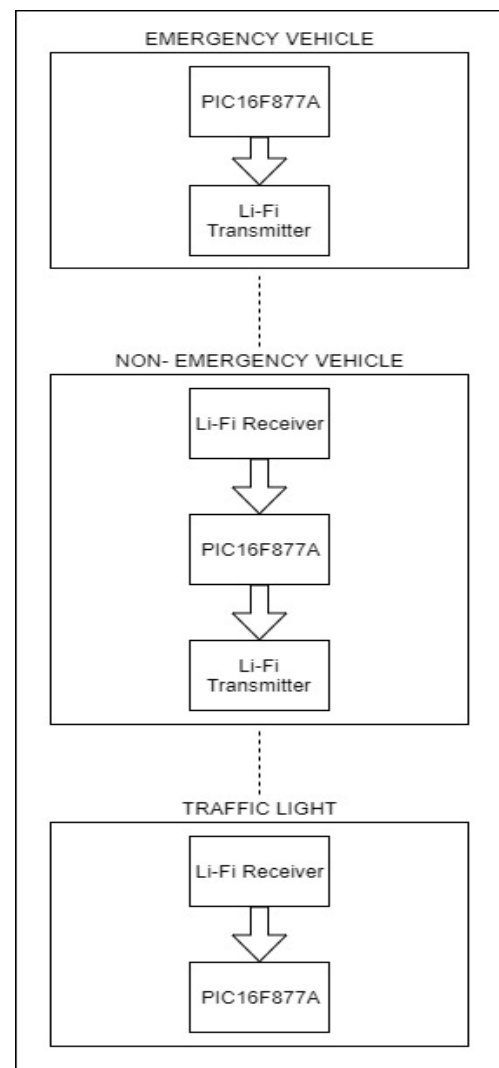


Fig. 1. Block Diagram of Li-Fi based Traffic control system

1) *PIC16F877A*

PIC16F877A was used to implement Li-Fi transceiver module using an LED and a photodiode. It is a low-cost, low power consuming microcontroller [10] [11]. It has a UART module which supports baud rate from 0.3 kbps to 57.6 kbps. In this paper, 2.4 kbps was chosen as the data rate of the Li-Fi communication system. Fig. 2, shows the block diagram of Li-Fi transceiver module.

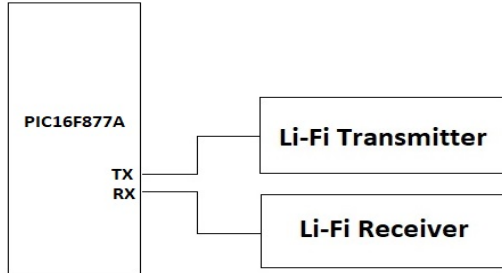


Fig. 2. Block Diagram of Li-Fi transceiver using PIC16F877A

2) *Li-Fi Transmitter*

In the circuit diagram of a basic Li-Fi transmitter shown in Fig. 3, a NPN transistor, BC547, is used as a current amplifier in CE configuration. Other types of current amplification circuits or signal power amplifying circuits can be used depending upon the LED lamp to be driven. STP55NF06 is a N-channel power MOSFET used to modulate the intensity of the LED lamp as per the data stream at a data rate of 2.4 kbps. The UART transmitter pin of PIC16F877A is connected to the 'Tx' pin of Li-Fi transmitter module.

This module supports data transmitted at baud rates 1.2 or 2.4 kbps. Any LED based headlights which are available in market can be converted into a Li-Fi transmitter by connecting it with the circuit shown in Fig. 3.

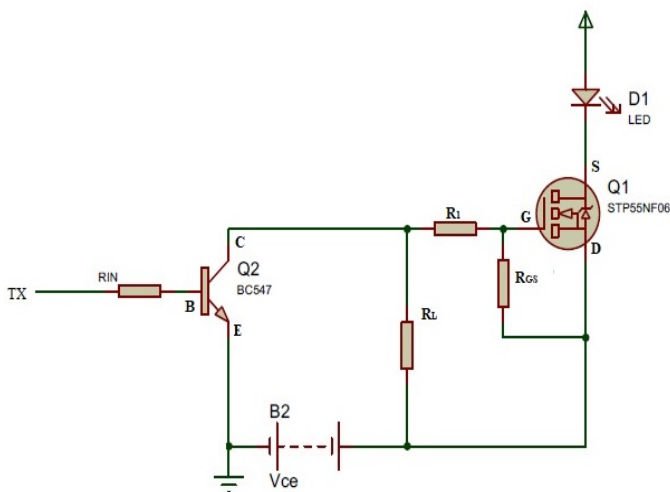


Fig. 3. Circuit diagram of Li-Fi transmitter module

3) *Li-Fi Receiver*

Fig. 4, shows the receiver circuit which consists of a photodiode, an operational amplifier LM358N, a potentiometer and resistors. The output of the photodiode is connected to the non-inverting terminal of the op-amp and the potentiometer is connected to the inverting terminal. By comparing the input voltage from the photodiode with the reference voltage set using the potentiometer, the op-amp will either set the output to the maximum positive voltage or maximum negative voltage. In this way, the op-amp differentiates between logic 1 and logic 0 of the transmitted bit stream.

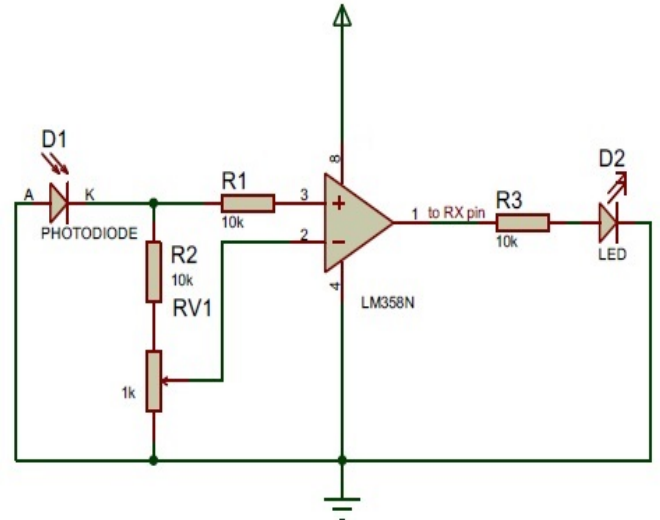


Fig. 4. Circuit diagram of a Li-Fi receiver module

IV. IMPLEMENTATION

Let us consider a scenario where an emergency vehicle has entered a lane which has many non-emergency vehicles ahead of it in a traffic-congested lane. Also, the traffic signal for the lane is red. In the current system the siren of the emergency vehicle is the only way to indicate its presence in a traffic-dense lane. Only if a traffic signal operator hears the siren the signal for the lane will be turned green. This system is time consuming and inefficient. Fig. 5, is the pictorial representation of the scenario considered.

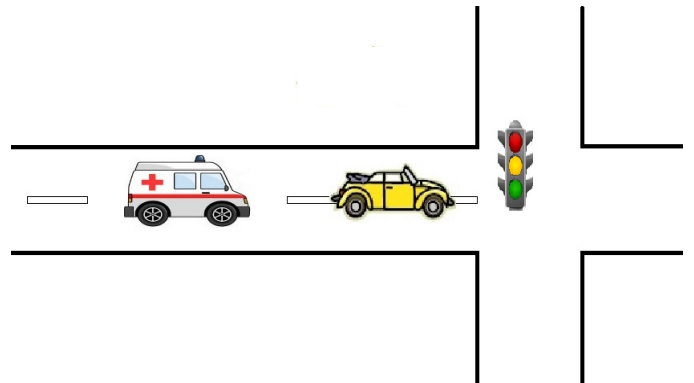


Fig. 5. Scenario with ambulance as emergency vehicle

Let us consider the similar scenario in which Li-Fi based automatic traffic signal control for emergency vehicles is implemented.

An emergency vehicle consists of a Li-Fi transmitter which is implemented using its headlight. When it enters a traffic-congested lane, it broadcasts an alert message like "EMERGENCY". If there is a non-emergency vehicle in front of it, the alert message will be received by the transceiver of the non-emergency vehicle. When the non-emergency vehicle receives the alert message, the driver of the vehicle will be informed using the vehicle infotainment system. At the same time, it will transmit the alert message either to a vehicle, if present in front or to the traffic signal control. The signal control, on receiving the alert message, will turn green.

The alert message can be hopped across any number of vehicles till it reaches the traffic signal control. Therefore, however long the traffic jam is, the alert message will reach the traffic signal control in a short span of time.

Fig. 6, shows the prototype developed to implement the scenario depicted in Fig. 5. This prototype shows how an alert message propagates from an emergency vehicle to a traffic signal controller through a non-emergency vehicle using Li-Fi as the communication protocol.

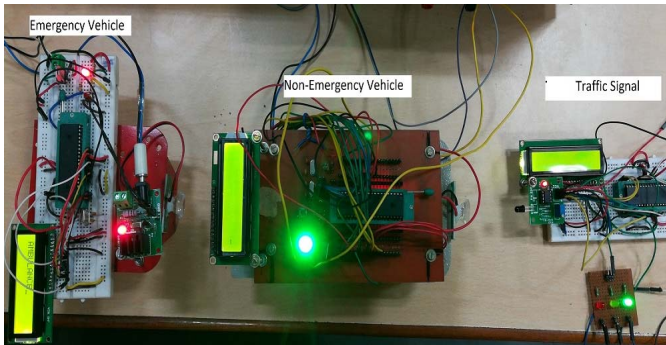


Fig. 6. Implementation of Li-Fi based automatic traffic control for emergency vehicles.

Fig. 7, shows the status of the traffic signal controller prototype before reception of any alert message. The signal is RED as it has not received the complete alert message. The alert message, in this case, is "AMBULANCE".

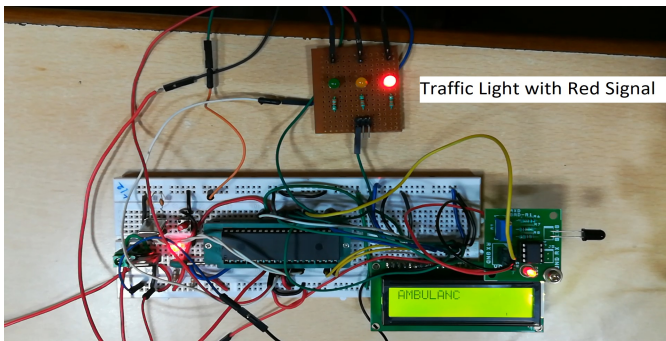


Fig. 7. Traffic light with Red signal before the alert message is received completely.

Fig. 8, shows status of the traffic signal prototype after the reception of the alert message. The signal has turned GREEN as it has received the complete alert message "AMBULANCE".

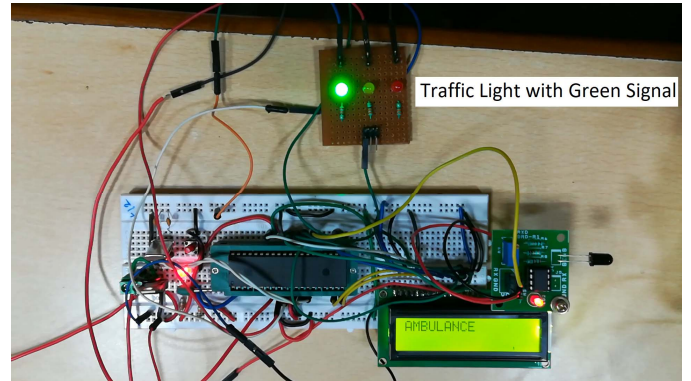


Fig. 8. Traffic light with Green signal after the alert message is received completely.

V. ADVANTAGES OF USING LI-FI IN THE TRAFFIC CONTROL SYSTEM

There are mainly four advantages in using Li-Fi in implementing this system. They are:

i. Capacity:

Li-Fi uses visible light spectrum for communication which is almost 10000 times than that of radio spectrum and most of the visible light spectrum is yet to be utilized.

ii. Availability:

In wireless data transmission, electromagnetic waves, mainly radio waves, are used for communication. There are 1.4 million radio base stations that are used for radio communication. But at the same time there are 1.2 billion vehicles on road. Therefore, in this system we can use the LED head and tail lights of any vehicle to carry out the data transmission and reception.

iii. Efficiency:

The average power consumption of a cellular base station is 2-6 kW out of which only 5 per cent is used for data transmission. On the other hand, the headlights used in commercial vehicles are rated as 3 W per light. Therefore, Li-Fi based traffic management system is far more superior when compared to cellular based traffic management system.

iv. Cost:

According to a study, to have 100 per cent coverage of cellular network in India, we would

have to spend at least 3150 million INR. On the other hand, cost of converting any LED light into a Li-Fi module would be less than 1000 INR. Even though Wi-Fi and ZigBee would cost approximately the same, the radio spectrum they utilize for communication is congested.

v. Safety:

Li-Fi communication uses visible light spectrum which has no harmful effect on the human body unlike Gamma rays, X-rays, Ultraviolet rays etc. which may even cause mutations at the cellular level.

VI. DISADVANTAGES OF USING LI-FI IN THE TRAFFIC CONTROL SYSTEM

Along with the advantages in using Li-Fi in this system there are some disadvantages associated in its usage.

- i. The vehicle head and tail lights need to be kept ON even during the day for Li-Fi to work but the intensity can be maintained at a minimum level.
- ii. Li-Fi works efficiently when the transmitter and receiver are placed in Line of Sight. Any deviation from this position can lead to miscommunication.
- iii. During the day, intense sunlight may interfere in Li-Fi communication. Also, other light sources, such as normal light bulbs, may also interrupt the communication.

VII. RESULTS

In this paper, a single hop scenario was implemented. The time taken for the alert message to propagate from an emergency vehicle to a non-emergency vehicle and then to the traffic signal control is approximately one second. This time can be further reduced by implementing the same functionality using optimized code. The time can also be reduced by designing better modulation and demodulation circuits which supports higher data rates. The alert message can also be encrypted to make it secure.

VIII. CONCLUSION

Using the Li-Fi communication infrastructure various other data can also be sent from one vehicle to another. Traffic density based automatic traffic signal control can be implemented. Highway toll collection, accident notification and other important traffic related parameters can also be implemented using the same model. If this system is implemented in the real world, many lives can be saved as the emergency vehicles would wait for very less time in traffic jams.

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